

June 7, 2026

Editor
Computers & Industrial Engineering

Dear Editor,

Please consider our manuscript, “A Reproducible Local-Ratio and SCC-Refinement Framework for Weighted Ordering in Directed Graphs,” for publication in *Computers & Industrial Engineering*.

The paper studies the minimum weighted feedback arc set problem on nonnegative weighted directed graphs through a reproducible computational methodology for weighted ordering. The manuscript contributes an engineered local-ratio topological add-back seed, a complementary weighted seed, and an incumbent-protected refinement over strongly connected components. The empirical study combines exact validation on small instances, component ablation, external baseline comparisons on sparse benchmark graphs, a public application-facing ordering example, and a dense transfer study on instances from the Linear Ordering Problem Library. These public datasets are used as reproducible surrogates for precedence-conflict resolution, dependency repair, and workflow-coordination settings where industrial data are often proprietary or lack standardized conflict weights. The resulting contribution is a scope-bounded heuristic framework: it is strongest on sparse nonnegative weighted digraphs, while the dense LOLIB study explicitly marks a regime where a matrix-based pairwise-ordering external method is stronger.

The submission package includes a double-anonymized review manuscript and is prepared to reference an anonymous reproducibility artifact for review. The work is original, has not been published previously, and is not under consideration elsewhere. The manuscript states that the study received no specific external funding, and it includes the required declaration regarding the use of AI-assisted tools in manuscript preparation.

Thank you for your consideration.

Sincerely,

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